A POLICY FRAMEWORK FOR COAL-ASSET REPURPOSING IN INDONESIA

Executive Summary

Indonesia, like several other Southeast Asian countries, is no exception to the economic dependence on coal. Around 67% of the country's electricity generation comes from coal-fired power plants, and coal mining accounts for 1.8% of the country's GDP between 2019 and 2020.[1] Yet the dependence is increasingly costly. The national electrical utility, PT Perusahaan Listrik Negara (PLN), remains locked into a system of ageing, complex, long-term contracts with independent power producers (IPPs) and underutilized coal-fired power plants. Indonesia is also extremely vulnerable to climate change, having problems such as deforestation and the displacement of people due to the coal mining industry. Despite a high stakeholder awareness and a solid legal foundation, large-scale coal asset repurposing has not materialized. PLN's limited implementation capacity and entrenched political will continue to hamper a just transition to alternative energy sources. This brief examines those key barriers to a just energy transition and assesses three policy recommendations for coal asset repurposing:

1. Strengthening public-private partnerships (PPPs) for asset retirement
2. Implementing blended finance mechanisms to bridge key financial gaps
3. Applying just transition principles, including worker reskilling and economic diversification, to the transition framework.

Political Economy Analysis

The Political Economy of Coal Lock-In

Understanding why Indonesia has struggled to translate its policy ambitions into repurposing coal assets requires moving beyond financial and technical explanations and examining Indonesia's development model, driven by resource extraction. The resource extraction model has roots in Indonesia's colonial legacies, continuing even after independence, often at the cost of local communities and environmental protection.[2] Various actors, including government actors, mining companies, local communities, environmental NGOs and think-tanks, the private sector and financial providers, and state-owned enterprises, influence the energy sector. These stakeholders often have competing interests and conflicting positions on the energy sector, particularly regarding the just transition.

At the government level, the country's energy policy is determined by the National Energy Council, which brings together seven ministries, including finance, transport, energy and mineral resources, forestry, etc.[3] State-owned enterprises (SOEs) have a significant stake in the energy sector, as Indonesia has a centralized energy sector, and PLN has a monopoly over electricity distribution and owns most of the generation capacity. IPPs often sell power to PLN through structured power purchase agreements (PPAs). The government provides significant subsidies to

PLN that prevent renewable energy developers and IPPs from entering the electricity market, due to PLN's low electricity prices.[4] The surge in coal mining is most noticeable from 2010 onwards after the mining law No 4/2009 replaced its 1967 predecessor, introducing a new mining framework. The decentralization of licenses, the issuance of coal mining concessions, and the mandate of local ownership of these concessions led to a rapid sell-off of land to mining companies. Given the mass issuance of concessions accompanied by poor oversight mechanisms, the coal industry has entrenched interests, including widespread corruption and patronage.[5]

A significant chunk of coal mining companies that operate in Indonesia are domestic, and many of these companies are well-connected to the policymaking environment. The companies are family-owned and run by politico-business conglomerates. This has created fairly good connections between the coal industry and the political environment, possessing a powerful lobby and connections to politicians, military leaders, government, and international partners.[6] Booming coal production also saw international interest, particularly from China. China is now one of the leading funders of Indonesian coal power, with international investments totaling almost $118 million in coal infrastructure. There is a positive relationship between Chinese investments and Indonesian coal policies; Chinese investors partner with Indonesian coal stakeholders who lobby for policies favorable to coal infrastructure and investment.[7] Additionally, mining companies often have a fractured relationship with the local communities, who have health and livelihood-related concerns due to rampant coal mining. This issue has led to the development of several environmental NGOs focussed on protesting large-scale mining projects, raising awareness among indigenous communities of their rights, and documenting coal impacts to enable legal action.[8]

In this sense, Indonesia’s coal transition challenge reflects a deeper political-economy problem rather than a narrow energy-policy gap. Efforts towards a just transition and the repurposing of coal assets will continue to face resistance, as coal remains embedded in systems of state support and corporate influence. A just transition will depend on restructuring these power relations while centering the rights and livelihoods of affected communities.

Indonesia as a Planning Country

A useful typology for understanding where countries sit in the global just energy transition landscape is to categorize them into implementation, planning, and early-stage groups.[9] Implementation countries are primarily the United States and the European Union, which have moved beyond planning to funding economic diversification and phasing out coal. Planning countries, in contrast, have achieved stakeholder awareness of the need for a transition and are also in the process of building out national policies and participating in frameworks to facilitate the process. This bucket includes Indonesia, where every major stakeholder, from PLN to the national energy council and civil society organizations, is aware of the need for a transition. Yet as of 2025, actual coal retirements remain negligible. Finally, there are countries such as India and Vietnam where the transition is still a relatively new concept, requiring structured institutionalization and formal planning. In the case of Indonesia, it points to structural constraints that organizations such as PLN and the national government face in driving repurposing.[10]

Critically, Indonesia's planning deficit is most acute at the asset level: there is no standardized methodology for assessing which coal plants are viable candidates for repurposing versus full

retirement, nor any framework for what a repurposed site should look like in terms of land use, workforce transition, and community economic planning.

The JETP: Headline Ambition, Implementation Deficit

The Indonesia Just Energy Transition Partnership (JETP) was an agreement signed in 2022 to mobilize USD 20 billion in public and private financing to support a just energy transition in Indonesia.[11] The agreement aims to establish goals to achieve net-zero emissions in the power sector by 2050 and to accelerate renewable energy deployment, with renewable energy contributing at least 34% to power generation.[12] In practice, the JETP has struggled to convert pledges into projects. Of the total financing committed, only USD 3.1 billion has been approved as of late 2025, covering a limited set of programs and grants.[13] The challenges are deeper than disbursements, with the new administration pushing for coal expansion and promoting fossil fuels, leading to the 2023 Comprehensive Investment and Policy Plan's deprioritization of early coal retirement. Indonesia's policies and actions receive a highly insufficient rating from the Climate Action Tracker, driven by the country's growing off-grid coal power pipeline.[14] The cancellation in late 2025 of the planned early retirement of the Cirebon-1 power plant under the ADB's Energy Transition Mechanism is the most concrete illustration of this failure: Cirebon-1 was designed as the proof-of-concept for the entire ETM approach, and its cancellation signals that the institutional, legal, and financing conditions necessary for replication do not yet exist.[15] The Cirebon-1 failure showcased the absence of a site-specific just transition plan for the plant's workers and the surrounding community, which became a source of local resistance that complicated negotiations.

Key Barriers to Coal-Asset Repurposing

Despite a growing policy consensus and an increasingly unfavorable fiscal outlook for coal, Indonesia's just energy transition faces four interlocking barriers:

1. Economic Dependence and Labor Vulnerability

Coal is not simply an energy source; it is a livelihood system for hundreds of thousands of workers and regional economies. According to 2024 data from the Ministry of Energy and Mineral Resources, more than 267,000 individuals are employed in the coal mining industry. As many as 159,900 coal workers are in East Kalimantan, with the IEA designating this region as the most coal-dependent globally.[16] A study by CELIOS and Yayasan Cerah – two energy-focused nonprofits- found that shutting down even three coal power plants would result in at least 14,000 job losses and a total income loss of approximately USD 243 million. Yet the same study found that if those closures were accompanied by renewable energy development, they could generate over 639,000 new jobs and boost income by USD 5 billion.[17] As a dominant coal exporter, Indonesia relies on the revenue it earns, as it is an important source of foreign currency to finance large-scale national projects. Beyond the direct employment impacts, the economic case for coal repurposing is complicated by market distortions. Coal use is heavily incentivized in the domestic market through subsidies, specifically the domestic market obligation (DMO), which caps market prices at USD 70/ton, well below market prices, creating barriers to renewable energy adoption.[18]

Another barrier in this area is the market structure, which is quasi-monopolistic, with PLN being a key player. PLN is also bound by long-term PPA with IPPs, further prolonging the phase-out. Any coal asset repurposing framework must therefore begin at the site level, with a worker census, a skills mapping exercise, and a community economic diversification plan attached to each candidate plant before repurposing decisions are made.

2. Disproportionate Impacts on Women and People with Disabilities

Women and people with disabilities (PWD) face compounded vulnerabilities and are often underrepresented in the energy transition. Women and PWD are more likely to experience the effects of energy poverty and face barriers to accessing appropriate information on the transition. Research on inclusive energy transition in Indonesia is limited. Indonesia's Long-Term Strategy for Low Carbon and Climate Resilience emphasizes mainstreaming gender equality, disability, and social inclusion (GEDSI) through specific programs and budget allocations. Still, these are not adequately enforced by the Government.[19] Repurposing frameworks that fail to embed GEDSI principles from the design stage risk reproducing the same exclusions that characterized the coal economy, substituting one extractive system for another without meaningfully redistributing benefits.

3. Grid Infrastructure and Financing Gaps

Indonesia possesses substantial renewable energy potential, estimated at 3,692 GW by the International Renewable Energy Agency (IRENA). Despite this, the national renewable energy share reached only 16% by 2025, falling short of the 23% target.[20] The implementation of adequate storage systems and a smart grid is essential for managing renewable energy generation and ensuring efficient transmission. The country's geographical complexity, characterized by mountainous regions and an archipelagic layout, presents significant barriers to grid infrastructure and separates demand from supply centers.[21] Additionally, limited land availability further complicates renewable energy development, preventing the construction of large centralized RE stations. The transmission system is fully owned and built by PLN, so grid development largely depends on the organization's financial capabilities.[22]

The financing picture compounds the infrastructure challenge, as achieving a just energy transition in Indonesia would require investments of billions to trillions of US dollars. Although JETP was a promising financing avenue, many commitments from developed countries remain unrealized. Uncertainty has increased because the ministry coordinating with the JETP secretariat was dissolved in 2024. Private capital has not filled this gap, partly because Indonesia's 6.25 percent interest rate is higher than most developing markets, and the high credit risk and banking capital requirements increase the cost of financing renewable energy projects.[23]

Repurposing coal plant sites for solar, battery storage, or green hydrogen offers a partial solution to this infrastructure problem. Existing grid connections, land, and water access at CFPP sites represent stranded assets that can be redeployed rather than abandoned, reducing the capital cost of renewable buildout.[24]

4. Weak Regulatory Frameworks and Coordination Failures

The fourth barrier is institutional. Indonesia's regulatory framework for the energy transition remains underdeveloped, and coordination mechanisms across government ministries are fragmented. Scholars have identified a lack of coherence between national and regional bodies on energy policies, as well as frequent policy changes, which together exacerbate uncertainties surrounding the transition.[25] Land acquisition challenges, bureaucratic obstacles, and prolonged permitting processes pose significant barriers to renewable energy projects, creating an unstable environment for investors. Developers frequently cite the need for greater transparency in tender processes and improved alignment between national and regional regulations. At the national level, fragmentation of responsibility for energy transition and national energy policy development across multiple ministries impedes effective implementation. At the regional level, local governments often rely on national-level actions.[26] For example, the Job Creation Law enacted in 2020 removed local governments' authority to set electricity tariffs, thereby limiting their capacity to incentivize renewable energy development. Additionally, planning for renewable energy sources is the responsibility of PLN, which limits local governments' ability to install renewables only in off-grid communities.[27]

Taken together, these four barriers reflect the structural conditions of Indonesia's coal system, embedded in its political economy and governance architecture. Section 4 examines three policy recommendations that can engage all four simultaneously.

Policy Recommendations

1. Strengthening Public-Private Partnerships (PPPs) For Early Retirement of CFPPs

A politically viable pathway for coal asset repurposing in Indonesia lies in leveraging PPPs to retire and convert CFFPs, without requiring PLN to pursue full divestment or absorb huge upfront costs in the transition. A PPP is a contractual arrangement between the public (PLN and the national government) and private entities through which the risks and rewards of large-scale projects are shared, promoting innovation and value-for-money service delivery.[28] This method is particularly suited to Indonesia, as it enables the state to retain strategic control over grid infrastructure while allowing the private developer to use its technical expertise to design and operate the infrastructure.

PPPs are already being leveraged by the Indonesian Government for its de-dieselization programs, with private developers co-developing clean energy such as solar PV alongside PLN. Legal frameworks for PPPs also exist, with Presidential Regulation No. 38 of 2015 enabling their use in infrastructure procurement. For PLN-owned CFFPs, PPP arrangements bring in private developers to finance renewable energy development, such as solar and battery storage, enhancing grid stability and flexibility in the region.[29]

Existing PPP frameworks, however, have a few structural limitations: PLN typically holds 51% ownership, reducing the return on investment for private developers and making such partnerships less attractive. A solution to this is the recently established sovereign wealth fund: Danantara. Danantara was established to restructure and optimize SOEs and identify energy transition as a

key priority. It is therefore uniquely positioned to act as a coordinating body between PLN and private developers in PPPs. The organization can aid in finding eligible assets, establishing frameworks and compliance mechanisms for PPPs, and ensuring repurposing pathways embody just-transition principles.[30]

2. Enhancing Blended Finance Mechanisms for Coal Asset Repurposing

The barriers identified in section 3, including high capital costs, low investor participation, and slow disbursements of JETP commitments, point to weaknesses in Indonesia's financial architecture for energy transitions. Addressing these requires a collaborative effort between philanthropists, private investors, the public sector, and multilateral development banks. Blended finance mechanisms are strategies that incentivize private-sector engagement in projects that deliver positive social and environmental outcomes. They are an effective way to de-risk renewable energy projects for investors by establishing risk-sharing mechanisms, creating partnerships among diverse bodies, and helping build policy frameworks that encourage renewable energy adoption.[31]

The ADB's Energy Transition Mechanism was launched as a platform for blended financing to accelerate the retirement of CFPPs. Despite the cancellation of Cirebon-1, it remains one of the most developed instruments for this purpose in the Indonesian context. A key lesson from that failure is that blended finance approaches must be reframed as economic development opportunities, linking repurposing projects explicitly to national priorities such as job creation and fiscal savings. At the transactional level, concessional loans and grants from multilateral development banks and philanthropic capital can be deployed to cover stranded asset costs associated with early PPA termination, which represents a large financial barrier to IPP-owned CFPP retirement.[32] Existing capital facilities, including those established by the ADB and the Global Energy Alliance for People and Planet, can strengthen technical expertise and build implementation capacity across the ecosystem, making individual projects more bankable. Energy transition platforms (ETPs) can further support this process by partnering with development finance institutions such as the ADB, IFC, and the World Bank to provide transaction advisory services and feasibility study support, two pain points that currently prevent private capital from entering the Indonesian repurposing market. By covering transaction costs and collaborating with the Indonesian government on enabling regulatory frameworks, ETPs can expand the supply of investable projects and build the knowledge-sharing infrastructure the market currently lacks.[33]

Critically, blended finance must not be designed as purely financial instruments. For the energy transition to be just, it must consider all socioeconomic risks and opportunities and address them for affected stakeholders. This includes preventive actions, such as reskilling programs, to help workers avoid losing their jobs, as well as transformative actions to create green jobs and diversify local economies. Financing agreements should therefore include explicit social conditionalities, such as worker-transition plans and community consultation processes at each repurposing site, to ensure that affected communities are not left behind in the energy transition.[34]

3. Embed Just Transition Principles as Conditionalities Within Coal Asset Repurposing Frameworks

The third recommendation addresses whether workers and vulnerable groups most affected by coal dependence are treated as active participants in the just transition process. A just transition planning layer must be built into every repurposing agreement from the beginning. Before a CFPP plant is approved for repurposing, a site-level just transition plan must be completed and publicly disclosed. Drawing on the Climate Policy Initiative's framework for quantifying just transition investment needs, this plan should identify preventive actions, including reskilling programs to prevent workers from losing their jobs and debt refinancing to avoid significant financial losses from early retirement.[35] Beyond jobs and reskilling programs, the plan should focus on the broader ecosystem (the number of informal jobs and local economic diversification). Stakeholders in coal-dependent regions have identified afforestation, floating photovoltaic installations on former mining sites, agroforestry systems, and ecotourism as viable alternatives to fill the revenue gap left by coal closures. These solutions must be explored with the affected communities through skill assessments and the establishment of adequate infrastructure to transition to such activities.[36]

Pai et al. (2021), in their case study of Jharkhand, find that coal dependence extends far beyond formal employment to include indirect and informal workers, pensioners, and entire community supply chains that have not been adequately quantified, indicative that transition planning built around workforce reskilling alone will be insufficient without a broader understanding of the coal ecosystem and investment in regional economic diversification.[37] This finding reinforces the need for Indonesia to move beyond generic workforce training and invest in site-specific economic diversification plans. A good example of this is the Sawahlunto region in Indonesia, a former coal-mining town that was transformed into a UNESCO-recognized tourism destination. The old mines and railway infrastructure support culture-based tourism. The successful transition was due to proactive early planning that included stakeholders from mining companies, local government, and the community.[38] An additional point to recognize is that reskilling and green jobs creation programs should include participation quotas and accessibility for women and PWD to ensure they have a greater stake in the energy sector and transition processes. MEMR Regulation No. 10/2025, which establishes the energy transition roadmap for Indonesia, has a narrow focus on metrics related to coal power, economic value, and emissions reductions, and fails to address worker and inclusivity-related issues adequately. This regulation should be revised to mandate GEDSI assessments, worker transition plans, and community consultation processes as prerequisites for CFPP retirement and repurposement.[39]

Conclusion

This policy brief examined key hurdles to a just energy transition and to the repurposing of coal assets in Indonesia. As evident throughout, a just energy transition requires more than environmental commitments; it is a complex political process, particularly for developing countries grappling with socio-economic and development challenges. The barriers examined here, including entrenched coal interests, monopolistic market structures, weak regulations, and

the exclusion of vulnerable communities from decision-making, are not unique to Indonesia. They reflect broader challenges across coal-dependent economies of the Global South, where international commitments increasingly compress transition timelines. At the same time, the governance institutions and political conditions required for change remain underdeveloped.

To accelerate Indonesia's coal retirement strategy, a hybrid approach that combines PPP investment, blended finance mechanisms, and economic diversification plans tailored to each CFPP site makes a credible just transition more likely. When these approaches are paired with clear regulatory frameworks, alignment across ministries and levels of government, and government support, clean and equitable energy transitions become achievable. Danantara's role as the coordinating body for state asset management can be pivotal in translating these frameworks from aspirational commitments into executable policies.

The success of Indonesia's transition will be measured not only by the gigawatts retired but also by whether the workers of East Kalimantan, the women excluded from energy decision-making, and coal-dependent communities are centered in the transition plans.

END NOTES

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